Alternator Rotational frequency in RPM (Revolutions per minute):

ν *is the rotational frequency*, $[rpm]$

$$\nu \triangleq \frac{revolution}{second} = \frac{1}{s} = 1\,Hz = frequency \Rightarrow frequency \cdot \frac{60\text{s}}{60\text{s}} = \frac{frequency \cdot 60\text{s}}{minute} = [rpm] \Rightarrow$$

$$\nu \triangleq \frac{revolution}{minute} = \frac{f \cdot 60\text{s}}{min} = [rpm]$$

n_{pp} *is the number of pole pairs* $\Rightarrow 6$

n_p *is the number of poles* $\Rightarrow 2 \cdot n_{pp} = 12$

1 *Revolution* $\triangleq 6\text{pp} = 12\text{p}$

f_{pp} *is the frequency of one pole pair*, $[Hz]$

f_{rpp} *is the frequency of all pole pairs for* 1 *revolution*, $[Hz]$

$$f_{rpp} = \frac{f_{pp}}{n_{pp}}$$

$$\nu = f_{rpp} \cdot 60\text{s} = \frac{f_{pp}}{n_{pp}} \cdot 60\text{s} \Rightarrow f_{pp} = \frac{\nu \cdot n_{pp}}{60\text{s}}$$

Rotating Magnetic Field (RMF) Rotational frequency in RPM (Revolutions per minute):

n_{pp} *is the number of pole pairs* $\Rightarrow 2$

n_p *is the number of poles* $\Rightarrow 2 \cdot n_{pp} = 4$

n_{Cpp} *is the number of Coils for one pole pair* $\Rightarrow 9$

n_{Cp} *is the number of Coils for all poles for* 1 *RMF Revolution* $\Rightarrow 2 \cdot n_{Cpp} = 18$

η_{pp} *is the factor difference for the pole pairs* $\Rightarrow 3$

1 *RMF Revolution* $\triangleq 2\text{pp} = 4\text{p}$

$\eta_{pp} \cdot$ *RMF Revolutions* $\triangleq 1$ *Alternator Revolution* $\triangleq 3 \cdot 2\text{pp} = 6\text{p}$

f_{RSR} *is the clock frequency of the Ring Shift Register*, $[Hz]$

f_{Cp} *is the frequency of all coils for* 1 *RMF revolution*, $[Hz]$

$$f_{Cp} = \frac{f_{RSR}}{n_{Cp}}$$

ν_{RMF} *is the rotational frequency of the RMF*, $[rpm]$

$$\nu_{RMF} = f_{CP} \cdot 60\text{s} = \frac{f_{RSR}}{n_{Cp}} \cdot 60\text{s} \Rightarrow f_{RSR} = \frac{\nu_{RMF} \cdot n_{Cp}}{60\text{s}}$$

ν_{ALT} *is the Alternators equivalent rotational frequency of the RMF*, $[rpm]$

$$\nu_{ALT} = \frac{\nu_{RMF}}{\eta_{pp}} = \frac{f_{CP} \cdot min}{\eta_{pp}} = \frac{f_{RSR}}{n_{Cp}} \cdot \frac{min}{\eta_{pp}} \Rightarrow f_{RSR} = \nu_{ALT} \cdot n_{Cp} \cdot \frac{\eta_{pp}}{min}$$